AGENDA

- We will discuss:
  - Linear Regression
  - Polynomial Regression
  - Ridge Regression (for regularization)
  - Evaluation Metrics

At the end of this lecture, you should be able to:

- ✓ Explain the working principles
- ✓ Apply the techniques to solve regression problems.
- ✓ Evaluate the performance of a regression task

Linear Regression

Linear Regression (Scalar Function)

- **Linear Regression** learns a model which is a **linear combination of features** of the input example.

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$,

where $\mathbf{x}$ is a d-dimensional feature vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_d \end{bmatrix}$$

Output: $f_{\mathbf{w}}(\mathbf{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_d x_d = \mathbf{x}^T \mathbf{w}$

where $\mathbf{w}$ is a d-dimensional vector of **parameters**

$$\mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix}$$

e.g., House Price Prediction

No.	Size	No. rooms	Location	Sold Price
1	112	2	648310	600000
…				
90	100	4	129580	801256

Linear Regression (Scalar Function)

➢ Learning (Training)

Find the optimal value $\mathbf{w}^*$ which **minimizes**:

$$J(\mathbf{w}) = \sum_{i=1}^{N} \boxed{(f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2}$$

Squared error loss

Least Squares Regression

- **Objective function**: the expression to be minimized or maximized

- **Loss function**: a measure of the difference between predicted output $f_{\mathbf{w}}(\mathbf{x}_i)$ and actual/desired/target output y_i.

$$\sum_{i=1}^{N} (f_{\mathbf{w}}(\mathbf{x}_i) - y_i)^2 = (\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y})$$

where $\mathbf{X} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,d} \\ \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,d} \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$

Linear Regression (Scalar Function)

➢ Learning (Training)

$$\underset{\mathbf{w}}{\operatorname{argmin}} \; (\mathbf{X}\mathbf{w}-\mathbf{y})^T(\mathbf{X}\mathbf{w}-\mathbf{y})$$

$$J(\mathbf{w}) = (\mathbf{X}\mathbf{w}-\mathbf{y})^T(\mathbf{X}\mathbf{w}-\mathbf{y}) = \left(\mathbf{w}^T\mathbf{X}^T-\mathbf{y}^T\right)(\mathbf{X}\mathbf{w}-\mathbf{y})$$

$$= \left(\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}-\mathbf{w}^T\mathbf{X}^T\mathbf{y}-\mathbf{y}^T\mathbf{X}\mathbf{w}+\mathbf{y}^T\mathbf{y}\right) = (\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}-2\mathbf{y}^T\mathbf{X}\mathbf{w}+\mathbf{y}^T\mathbf{y})$$

The optimal value $\mathbf{w}^*$ should have **zero gradient:**

$$\frac{dJ(\mathbf{w})}{d\mathbf{w}} = \mathbf{0}$$

Why?

$$\frac{d(\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}-2\mathbf{y}^T\mathbf{X}\mathbf{w}+\mathbf{y}^T\mathbf{y})}{d\mathbf{w}} = \mathbf{0}$$

$$\frac{d(\mathbf{w}^T\mathbf{A}\mathbf{w})}{d\mathbf{w}} = (\mathbf{A}+\mathbf{A}^T)\mathbf{w}$$

$$\frac{d(\mathbf{y}^T\mathbf{A}\mathbf{w})}{d\mathbf{w}} = \mathbf{A}^T\mathbf{y}$$

$$2\mathbf{X}^T\mathbf{X}\mathbf{w}-2\mathbf{X}^T\mathbf{y} = \mathbf{0}$$

$$\mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

If $\mathbf{X}^T\mathbf{X}$ is invertible, $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

Linear Regression (Scalar Function)

➢ Learning (Training)

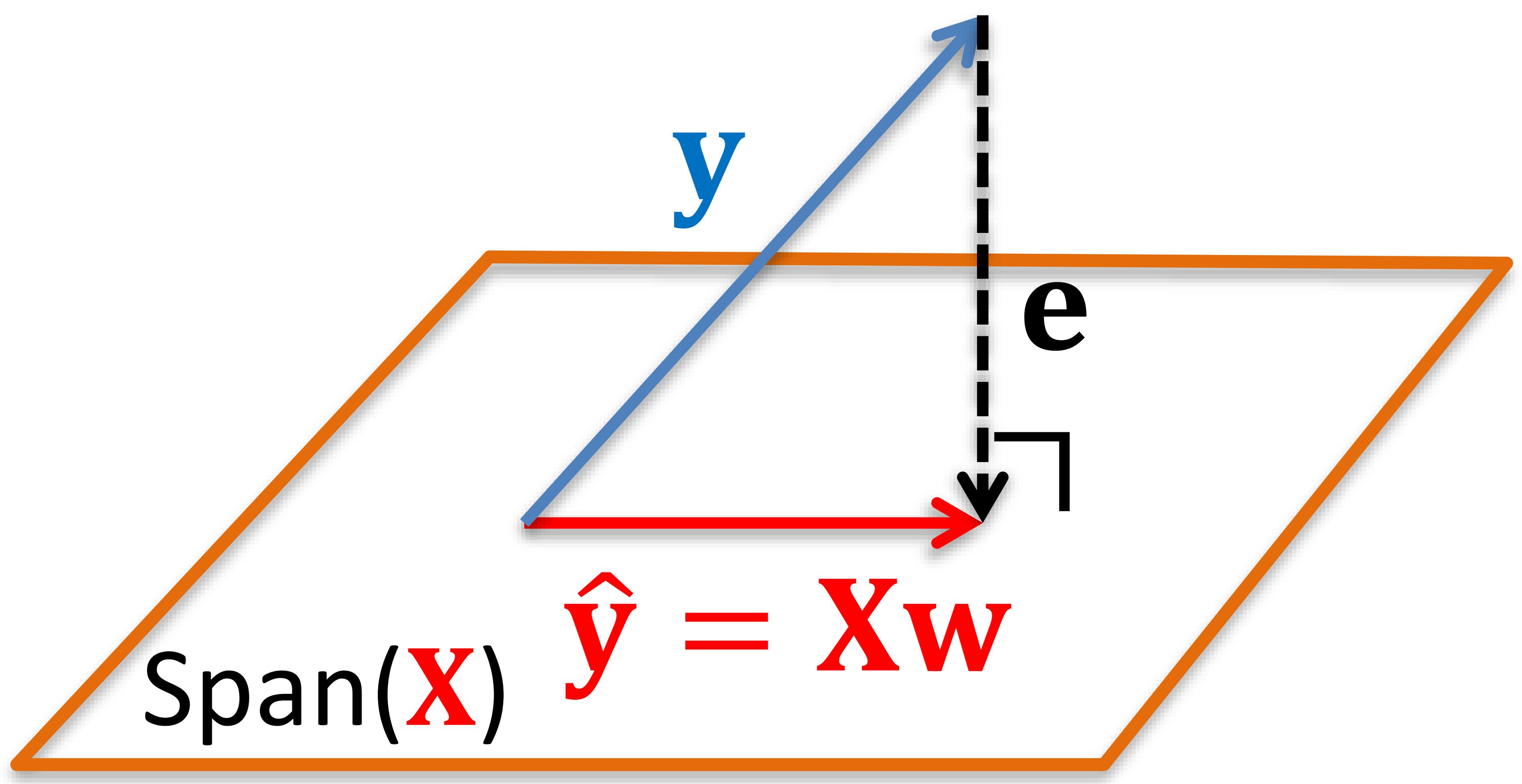
- Geometric Interpretation of Least Squares Solution

$$\mathbf{X}^T(\underbrace{\mathbf{X}\mathbf{w}-\mathbf{y}}_{\mathbf{e}}) = \mathbf{0}$$

$$\begin{bmatrix} x_{1,1} & \cdots & x_{N,1} \\ \vdots & \ddots & \vdots \\ x_{1,d} & \cdots & x_{N,d} \end{bmatrix} \mathbf{e} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\mathbf{e}$ is perpendicular to each row of $\mathbf{X}^T$ (i.e., each column of $\mathbf{X}$)

$\mathbf{e}$ is perpendicular to the column space of $\mathbf{X}$ (i.e., Span($\mathbf{X}$))



Linear Regression (Scalar Function)

➢ Prediction (Testing)

$$\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_{new}) = \mathbf{X}_{new}\mathbf{w}^* = \begin{bmatrix} x_{1,1}^{new} & \cdots & x_{1,d}^{new} \\ \vdots & \ddots & \vdots \\ x_{m,1}^{new} & \cdots & x_{m,d}^{new} \end{bmatrix} \begin{bmatrix} w_1^* \\ \vdots \\ w_d^* \end{bmatrix}$$

$$= \begin{bmatrix} (\mathbf{x}_1^{new})^T \mathbf{w}^* \\ \vdots \\ (\mathbf{x}_m^{new})^T \mathbf{w}^* \end{bmatrix}$$

Is the obtained $\mathbf{w}^*$ a global minimum?

Linear Regression (Scalar Function)

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7,9\}$ using least squares regression.

Step 1: Learning (work out $\mathbf{w}^*$)

$$\mathbf{X} = \begin{bmatrix} -10 \\ -8 \\ -3 \\ -1 \\ 2 \\ 8 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Since $\mathbf{X}^T\mathbf{X} = [242]$ is invertible

$$= [242]^{-1}[-10,-8,-3,-1,2,8]\begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$= -0.3512$$

Step 2: Prediction

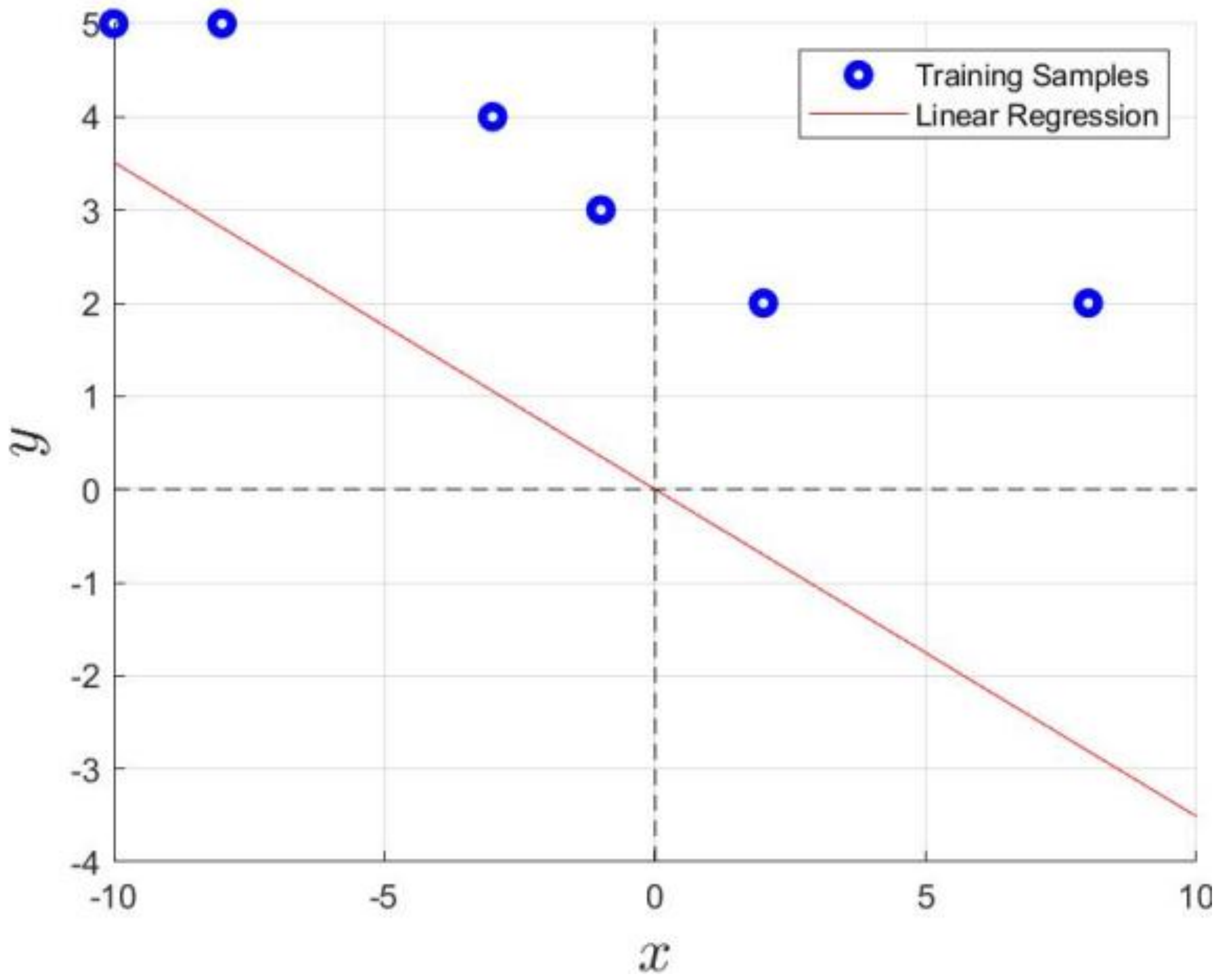
$$\mathbf{X}_t = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

$$\mathbf{y}_t = \mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t\mathbf{w}^* = \begin{bmatrix} 7 \\ 9 \end{bmatrix}[-0.3512]$$

$$= \begin{bmatrix} -2.4587 \\ -3.1612 \end{bmatrix}$$

Linear Regression (Scalar Function)

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.



$$f_{\mathbf{w}}(\mathbf{x}) = w_1 x_1 = -0.3512 x_1$$

~~The line always passes through the origin.~~

Linear Regression (Scalar Function)

➢ Linear Regression with bias/offset

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)$,

Output: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d = \mathbf{x}^T \mathbf{w}$

where $\mathbf{x}$ is a (*d*+*1*)-dimensional feature vector

$\mathbf{w}$ is a (*d*+*1*)-dimensional vector of parameters

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}$$

The bias/offset term w_0 is responsible for translating the line/plane/hyperplane away from the origin

Linear Regression (Scalar Function)

➢ Linear Regression with bias/offset

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$,

Output: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d = \mathbf{x}^T \mathbf{w}$

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix} \mathbf{w} = \begin{bmatrix} w_0 \\ \vdots \\ w_d \end{bmatrix}$$

-Learning: $\mathbf{w}^* = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$

where $\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \ldots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \ldots & x_{N,d} \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t \mathbf{w}^*$

where $\mathbf{X}_t = \begin{bmatrix} 1 & x_{1,1}^t & \ldots & x_{1,d}^t \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m,1}^t & \ldots & x_{m,d}^t \end{bmatrix}$, $\mathbf{w}^* = \begin{bmatrix} w_0^* \\ \vdots \\ w_d^* \end{bmatrix}$

Linear Regression (Scalar Function)

➢ Linear Regression with bias/offset

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7,9\}$ using least squares regression.

Step 1: Learning (work out $\mathbf{w}^*$)

$$\mathbf{X} = \begin{bmatrix} 1 & -10 \\ 1 & -8 \\ 1 & -3 \\ 1 & -1 \\ 1 & 2 \\ 1 & 8 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

$$\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Since $\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 6 & -12 \\ -12 & 242 \end{bmatrix}$ is invertible

$$= \begin{bmatrix} 6 & -12 \\ -12 & 242 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -10 \\ 1 & -8 \\ 1 & -3 \\ 1 & -1 \\ 1 & 2 \\ 1 & 8 \end{bmatrix}^T \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3.1055 \\ -0.1972 \end{bmatrix}$$

$$\mathbf{X}_t = \begin{bmatrix} 1 & 7 \\ 1 & 9 \end{bmatrix}$$

Step 2: Prediction

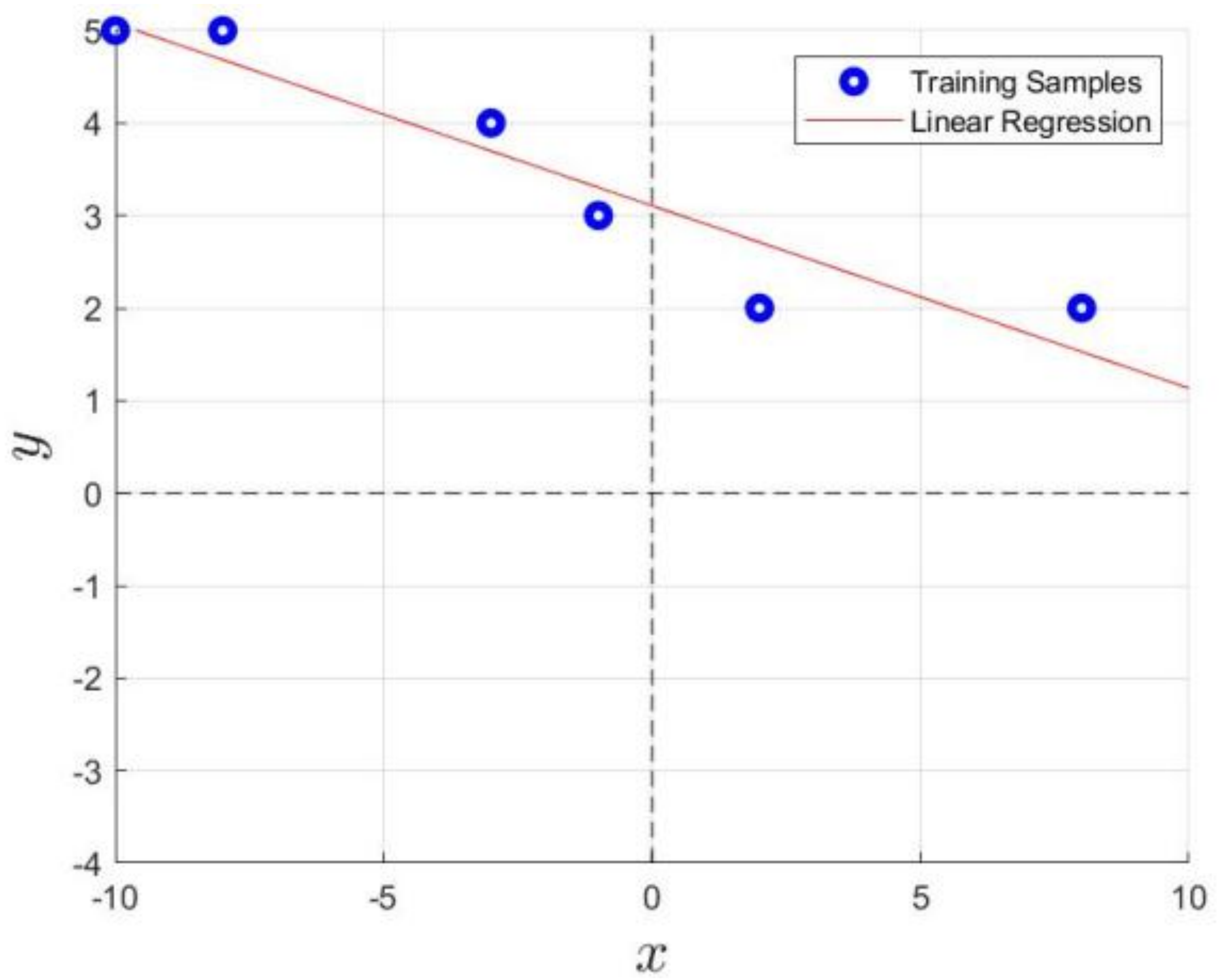
$$\mathbf{y}_t = \mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t\mathbf{w}^* = \begin{bmatrix} 1 & 7 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} 3.1055 \\ -0.1972 \end{bmatrix}$$

$$= \begin{bmatrix} 1.7248 \\ 1.3303 \end{bmatrix}$$

Linear Regression (Scalar Function)

➢ **Linear Regression** with bias/offset

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6}$
$= \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.

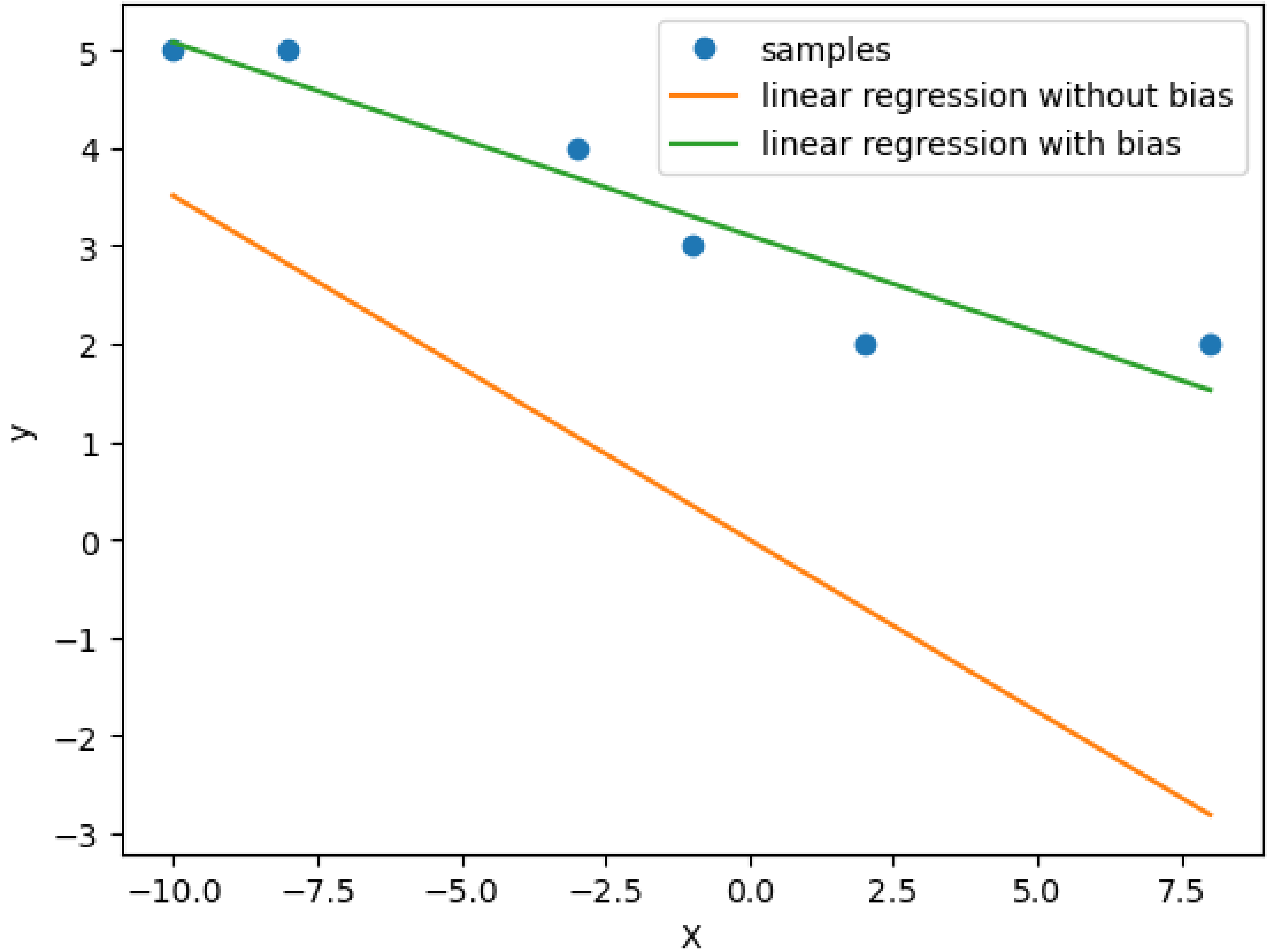


$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1$$
$$= 1.7251 + 1.3307 x_1$$

Linear Regression (Scalar Function)

➢ Comparison

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10, 5), (-8, 5), (-3, 4), (-1, 3), (2, 2), (8, 2)\}$, predict the test dataset $\{7, 9\}$ using least squares regression.



Linear Regression (Vector Function)

➢ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \ldots, (\mathbf{x}_N, \mathbf{y}_N)$,

where $\mathbf{x}$ is a *(d+1)*-dimensional feature vector

$\mathbf{y}$ is a *h*-dimensional target vector

$$\mathbf{x} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

e.g., predict multiple physiological signals

	Age	Weight	Physical activity level	Oxygen saturation	Heart rate	Blood pressure	Respiration rate
1							
…							

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{x}^T\mathbf{W}$, where **W** is a matrix of size (*d*+1)x *h*

$$\mathbf{W} = \begin{bmatrix} w_{0,1} & \cdots & w_{0,h} \\ \vdots & \ddots & \vdots \\ w_{d,1} & \cdots & w_{d,h} \end{bmatrix}$$

weight vector $\mathbf{w}_1$ for y_1 weight vector $\mathbf{w}_h$ for y_h

Equivalent to performing Linear Regression for each target output variable (h times)

$$\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = [f_{\mathbf{w}_1}(\mathbf{x}), \ldots, f_{\mathbf{w}_h}(\mathbf{x})]$$

Linear Regression (Vector Function)

➢ Learning

Find the optimal value $\mathbf{W}^*$ which minimizes:

$$J(\mathbf{W}) = \sum_{k=1}^{h}\sum_{i=1}^{N}\left(f_{\mathbf{w}_k}(\mathbf{x}_i) - y_{i,k}\right)^2$$

$$= \sum_{k=1}^{h}(\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)^T(\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} \end{bmatrix}$$

The optimal value $\mathbf{W}^*$ should have **zero gradient:**

$$\frac{dJ(\mathbf{W})}{d\mathbf{W}} = \begin{bmatrix} \boxed{\begin{matrix} \frac{\partial J}{\partial w_{0,1}} \\ \vdots \\ \frac{\partial J}{\partial w_{d,1}} \end{matrix}} & \begin{matrix} \cdots \\ \ddots \\ \cdots \end{matrix} & \boxed{\begin{matrix} \frac{\partial J}{\partial w_{0,h}} \\ \vdots \\ \frac{\partial J}{\partial w_{d,h}} \end{matrix}} \end{bmatrix} = \mathbf{0} \Rightarrow \frac{dJ(\mathbf{W})}{d\mathbf{w}_k} = \mathbf{0}, k = 1,2,\ldots,h$$

$$\frac{dJ(\mathbf{W})}{d\mathbf{w}_1} \qquad \frac{dJ(\mathbf{W})}{d\mathbf{w}_h}$$

$$\Downarrow$$

$$\mathbf{w}_k{}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}_k$$

Linear Regression (Vector Function)

➢ Learning

Find the optimal value for $\mathbf{W}^*$ which minimizes:

$$J(\mathbf{W}) = \sum_{k=1}^{h}\sum_{i=1}^{N}\left(f_{\mathbf{w}_k}(\mathbf{x}_i) - y_{i,k}\right)^2$$

$$= \sum_{k=1}^{h}(\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)^T(\mathbf{X}\mathbf{w}_k - \mathbf{y}_k)$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} \end{bmatrix}$$

$$\mathbf{W}^* = [\mathbf{w}_1{}^*, \ldots, \mathbf{w}_h{}^*] = [(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}_1, \ldots, (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}_h]$$

$$= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$$

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,h} \\ \vdots & \ddots & \vdots \\ y_{N,1} & \cdots & y_{N,h} \end{bmatrix}$$

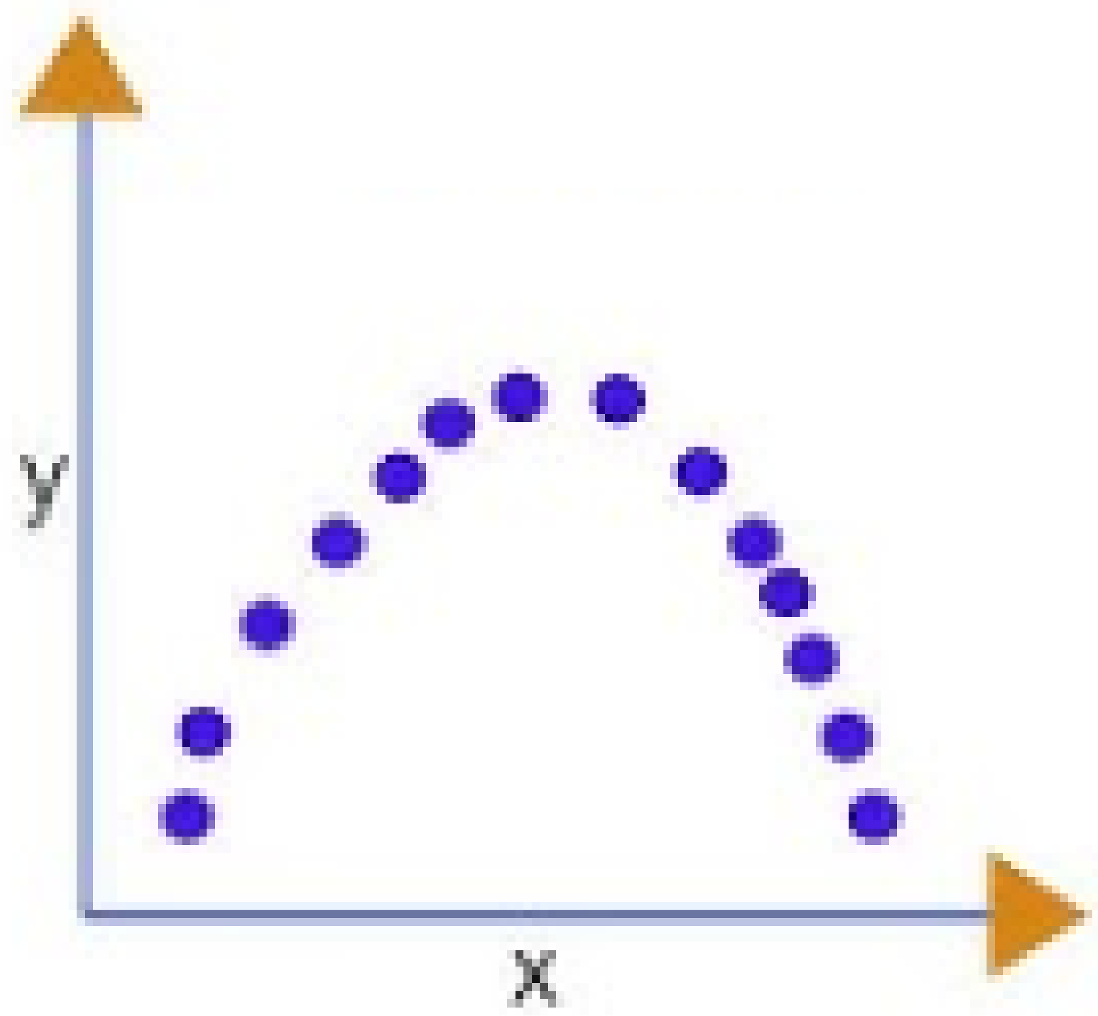
Is the obtained $\mathbf{W}^*$ a global minimum?

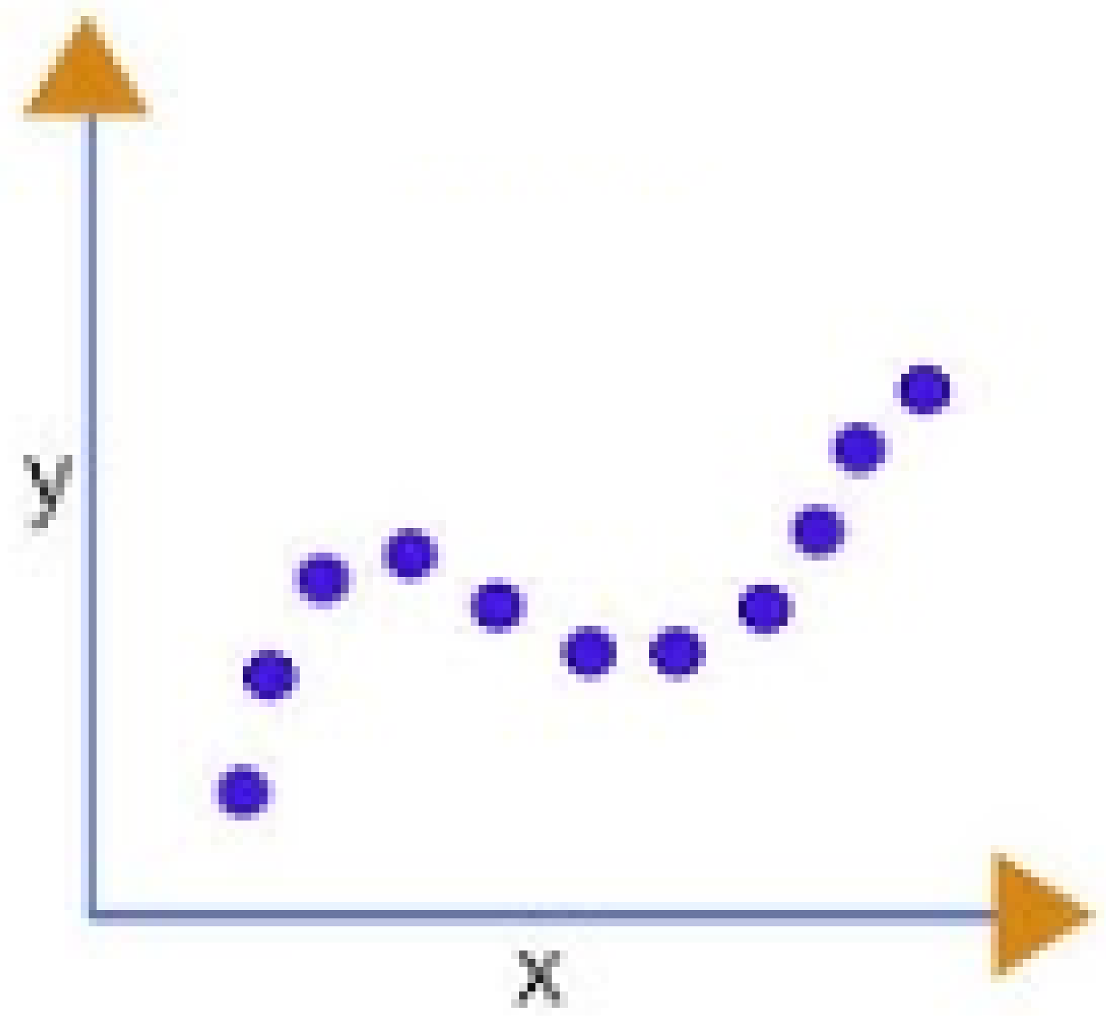
Linear Regression (Vector Function)

➢ Prediction

$$\mathbf{F}_{\mathbf{W}^*}(\mathbf{X}_t) = \mathbf{X}_t\mathbf{W}^* = \begin{bmatrix} 1 & x_{1,1}^t & \cdots & x_{1,d}^t \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{m,1}^t & \cdots & x_{m,d}^t \end{bmatrix} \begin{bmatrix} w_{0,1}^* & \cdots & w_{0,h}^* \\ \vdots & \ddots & \vdots \\ w_{d,1}^* & \cdots & w_{d,h}^* \end{bmatrix}$$

$$= \begin{bmatrix} (\mathbf{x}_1^t)^T\mathbf{w}_1^* & \cdots & (\mathbf{x}_1^t)^T\mathbf{w}_h^* \\ \vdots & \ddots & \vdots \\ (\mathbf{x}_m^t)^T\mathbf{w}_1^* & \cdots & (\mathbf{x}_m^t)^T\mathbf{w}_h^* \end{bmatrix}$$





Polynomial Regression

non-linear relationship between input and output

Polynomial Regression

➢ Polynomial Model

$$f_{\mathbf{w}}(\mathbf{x}) = \underbrace{\underbrace{\underbrace{w_0 + \sum_{i=1}^{d} w_i x_i}_{\text{1st order (Linear)}} + \sum_{i=1}^{d}\sum_{j=1}^{d} w_{ij} x_i x_j}_{\text{2nd order (Quadratic)}} + \sum_{i=1}^{d}\sum_{j=1}^{d}\sum_{k=1}^{d} w_{ijk} x_i x_j x_k}_{\text{3rd order (Cubic)}} + \cdots$$

Order number: Hyperparameter!

$w_0 + w_1 x_1 + w_{11} x_1 x_1$

$w_{12} x_1 x_2$

$w_{21} x_1 x_2$

e.g., d=1, 3rd order: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3$

d=2, 2nd order: $f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2 + w_{11} x_1^2 + w_{12} x_1 x_2 + w_{22} x_2^2$

No. parameters for n-th order polynomial model with d input variables?

$$\binom{n+d}{d} = \frac{(n+d)!}{n!\,d!}$$

No. parameters for *degree n terms* with d input variables?

$$\binom{n+d-1}{d-1} = \frac{(n+d-1)!}{n!\,(d-1)!}$$

- Proof: total No. paras for *n*-th order polynomial model with *d* input variables is $\binom{n+d}{d} = \frac{(n+d)!}{n!d!}$

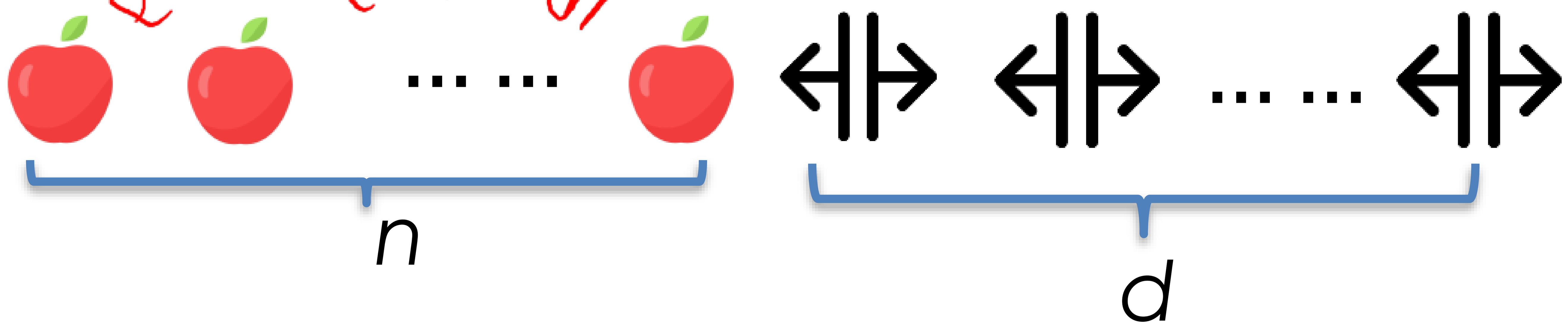
$$x_1^{k_1} x_2^{k_2} \dots x_d^{k_d}: k_1 + k_2 + \cdots + k_d \le n$$

$$\Updownarrow$$

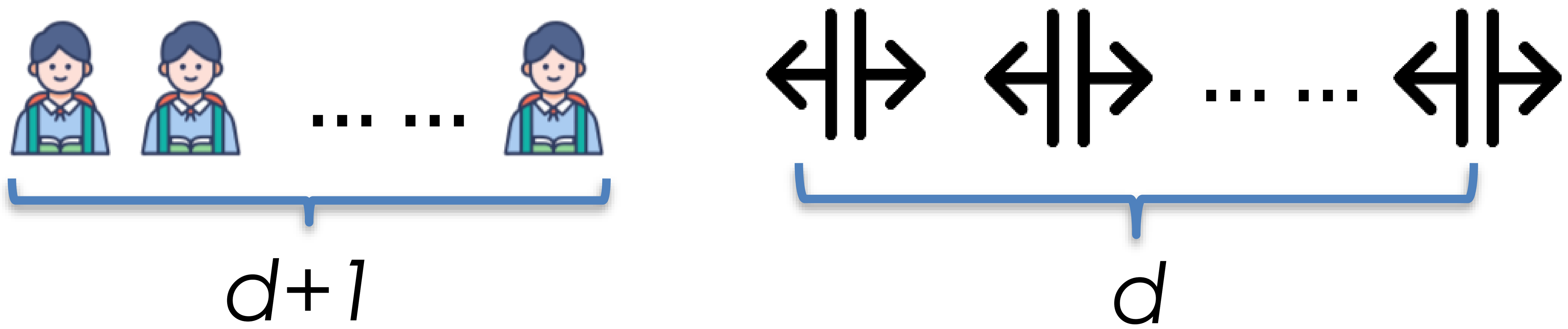
$$k_1 + k_2 + \cdots + k_d + k_{d+1} = n$$

Counting: equivalent to allocating *n* apples to *d*+1 students.

$\binom{n-1}{d}$



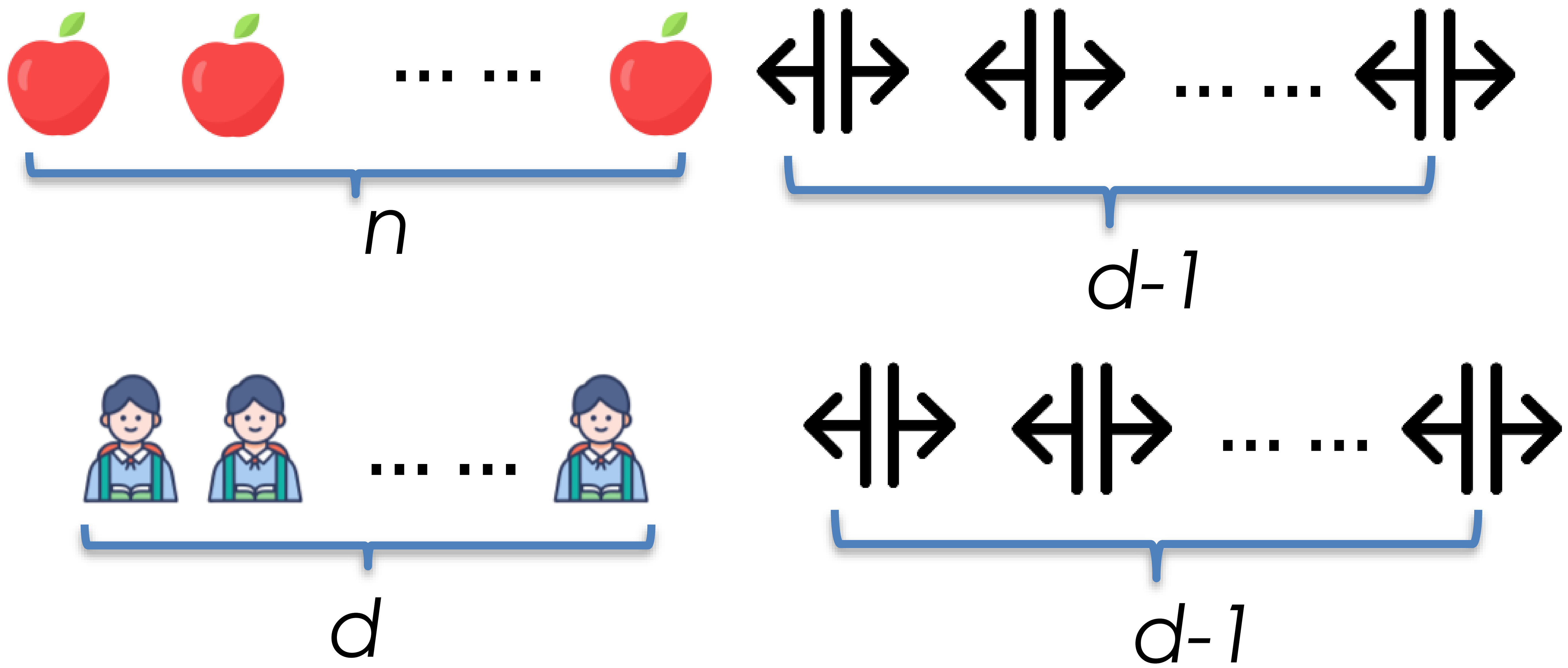
$$\binom{n+d}{d} = \binom{n+d}{n}$$



- Proof: No. paras for *degree n terms* with *d* input variables is $\binom{n+d-1}{d-1} = \frac{(n+d-1)!}{n!(d-1)!}$

$$x_1^{k_1} x_2^{k_2} \ldots x_d^{k_d}: k_1 + k_2 + \cdots + k_d = n$$

Counting: equivalent to allocating *n* apples to *d* students.



$$\binom{n+d-1}{d-1} = \binom{n+d-1}{n}$$

Polynomial Regression

➢ Polynomial Model

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + \sum_{i=1}^{d} w_i x_i + \sum_{i=1}^{d}\sum_{i=1}^{d} w_{ij} x_i x_j + \sum_{i=1}^{d}\sum_{i=1}^{d}\sum_{i=1}^{d} w_{ijk} x_i x_j x_k + \cdots$$

$$= f_{\mathbf{w}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T \mathbf{w}$$

$$\mathbf{p} = \begin{bmatrix} 1 \\ x_1 \\ \vdots \\ x_d \\ \vdots \\ x_i x_j \\ \vdots \\ x_i x_j x_k \\ \vdots \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \\ \vdots \\ w_{ij} \\ \vdots \\ w_{ijk} \\ \vdots \end{bmatrix}$$

Equivalent to:

polynomial transformation
+
linear regression

Polynomial Regression

➢ Learning

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Input: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \ldots, (\mathbf{x}_N, y_N)$,

Output: $f_{\mathbf{w}}(\mathbf{x}) = f_{\mathbf{w}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T\mathbf{w}$

Find the optimal value $\mathbf{w}^*$ which minimizes:

$$J(\mathbf{w}) = \sum_{i=1}^{N} (f_{\mathbf{w}}(\mathbf{p}(\mathbf{x}_i)) - y_i)^2 = (\mathbf{P}\mathbf{w} - \mathbf{y})^T(\mathbf{P}\mathbf{w} - \mathbf{y})$$

$$\mathbf{P} = \begin{bmatrix} 1 & x_{1,1} & \cdots & x_{1,d} & \cdots & x_{1,i}x_{1,j} & \cdots & x_{1,i}x_{1,j}x_{1,k} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{N,1} & \cdots & x_{N,d} & \cdots & x_{N,i}x_{N,j} & \cdots & x_{N,i}x_{N,j}x_{N,k} & \cdots \end{bmatrix}$$

Solution: $\mathbf{w}^* = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{y}$, If $\mathbf{P}^T\mathbf{P}$ is invertible

global minimum?

Polynomial Regression

➢ Prediction

$$\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{w}^*$$

$$= \begin{bmatrix} 1 & x_{1,1}^t & \cdots & x_{1,d}^t & \cdots & x_{1,i}^t x_{1,j}^t & \cdots & x_{1,i}^t x_{1,j}^t x_{1,k}^t & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{m,1}^t & \cdots & x_{m,d}^t & \cdots & x_{m,i}^t x_{m,j}^t & \cdots & x_{m,i}^t x_{m,j}^t x_{m,k}^t & \cdots \end{bmatrix} \begin{bmatrix} w_0^* \\ w_1^* \\ \vdots \\ w_d^* \\ \vdots \\ w_{ij}^* \\ \vdots \\ w_{ijk}^* \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} (\mathbf{p}_1^t)^T\mathbf{w}^* \\ \vdots \\ (\mathbf{p}_m^t)^T\mathbf{w}^* \end{bmatrix}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7,9\}$ using 3rd order polynomial regression.

Step 0: Polynomial Transformation ($\mathbf{X} \rightarrow \mathbf{P}$)

3rd order polynomial model with 1 input variable:

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3 = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_{11} \\ w_{111} \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 1 & x_{1,1} & x_{1,1}^2 & x_{1,1}^3 \\ 1 & x_{2,1} & x_{2,1}^2 & x_{2,1}^3 \\ 1 & x_{3,1} & x_{3,1}^2 & x_{3,1}^3 \\ 1 & x_{4,1} & x_{3,1}^2 & x_{4,1}^3 \\ 1 & x_{5,1} & x_{4,1}^2 & x_{5,1}^3 \\ 1 & x_{6,1} & x_{6,1}^2 & x_{6,1}^3 \end{bmatrix} = \begin{bmatrix} 1 & -10 & 100 & -1000 \\ 1 & -8 & 64 & -512 \\ 1 & -3 & 9 & -27 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \\ 1 & 8 & 64 & 512 \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} 5 \\ 5 \\ 4 \\ 3 \\ 2 \\ 2 \end{bmatrix}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7,9\}$ using 3rd order polynomial regression.

Step 1: Learning (work out $\mathbf{w}^*$)

Since $\mathbf{P}^T\mathbf{P}$ is invertible, $\mathbf{w}^* = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{y} = \begin{bmatrix} 2.6894 \\ -0.3772 \\ 0.0134 \\ 0.0029 \end{bmatrix}$

Step 2: Prediction

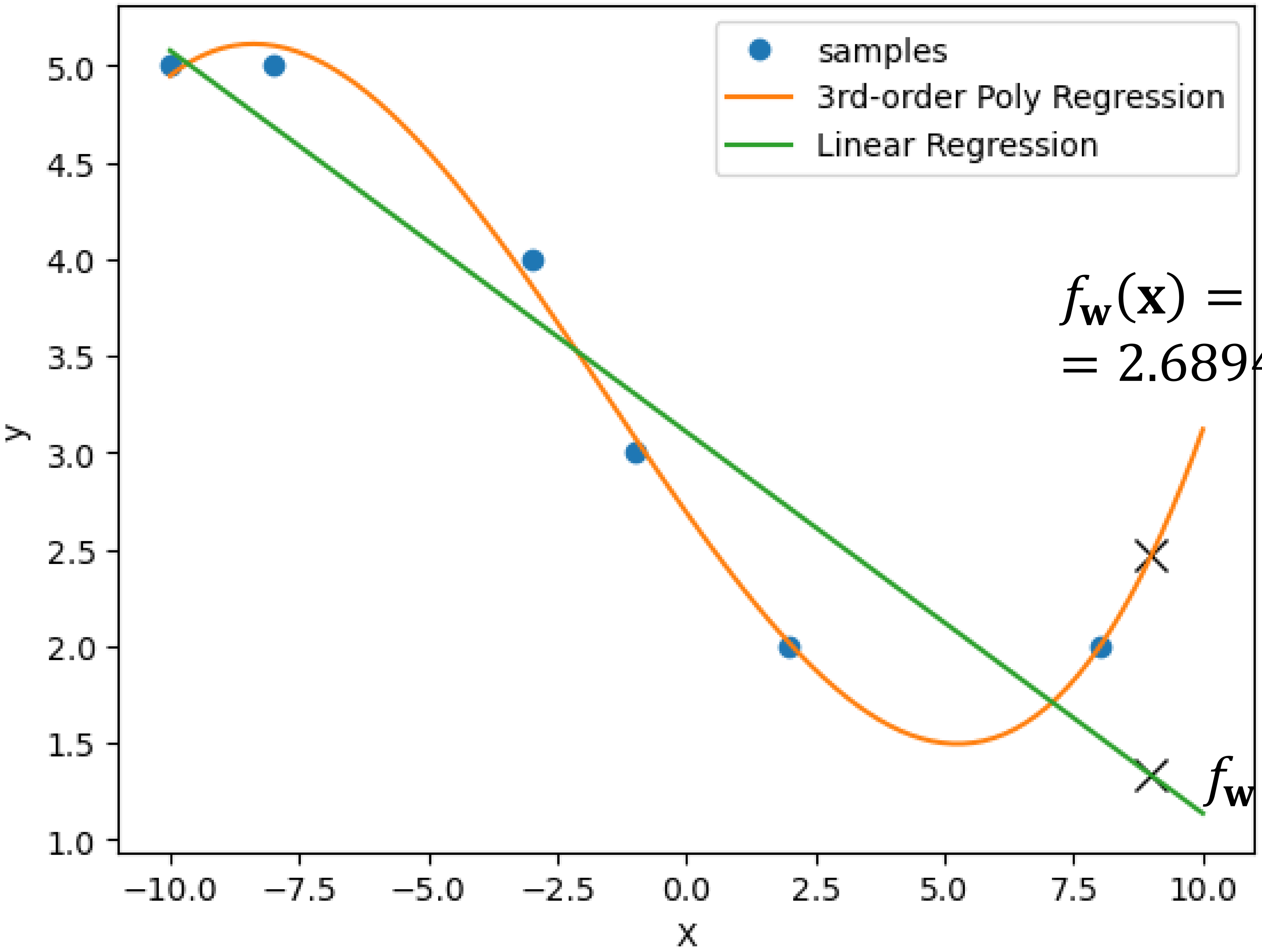
$$\mathbf{P}_t = \begin{bmatrix} 1 & x_{1,1}^t & (x_{1,1}^t)^2 & (x_{1,1}^t)^3 \\ 1 & x_{2,1}^t & (x_{2,1}^t)^2 & (x_{2,1}^t)^3 \end{bmatrix} = \begin{bmatrix} 1 & 7 & 49 & 343 \\ 1 & 9 & 81 & 729 \end{bmatrix}$$

$$\mathbf{y}_t = \mathbf{f}_{\mathbf{w}^*}(\mathbf{P}_t) = \mathbf{P}_t\mathbf{w}^* = \begin{bmatrix} 1 & 7 & 49 & 343 \\ 1 & 9 & 81 & 729 \end{bmatrix}\begin{bmatrix} 2.6894 \\ -0.3772 \\ 0.0134 \\ 0.0029 \end{bmatrix}$$

$$= \begin{bmatrix} 1.6874 \\ 2.4661 \end{bmatrix}$$

Polynomial Regression

Example: Given a training set $\{(x_i, y_i)\}_{i=1}^{6} = \{(-10,5),(-8,5),(-3,4),(-1,3),(2,2),(8,2)\}$, predict the test dataset $\{7,9\}$ using 3rd order polynomial regression.



$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 + w_{11} x_1^2 + w_{111} x_1^3$$
$$= 2.6894 - 0.3772 x_1 + 0.0134 x_1^2 + 0.0029 x_1^3$$

$$f_{\mathbf{w}}(\mathbf{x}) = w_0 + w_1 x_1 = 1.7251 + 1.3307 x_1$$

Polynomial Regression

➢ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N)$,

where $\mathbf{x}$ is a d-dimensional feature vector

$\mathbf{y}$ is a h-dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f}_{\mathbf{W}}(\mathbf{x}) = \mathbf{f}_{\mathbf{W}}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T\mathbf{W}$,

-Learning: $\mathbf{W}^* = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{W}^*$

Ridge Regression

If $\mathbf{X}^T\mathbf{X}$ or $\mathbf{P}^T\mathbf{P}$ is not invertible

Ridge Regression

➢ **Ridge Regression = Regression + L2 regularization**

Shrinks the regression coefficients w by imposing a penalty to their size.

Find the optimal value $\mathbf{w}^*$ which minimizes:

$$J(\mathbf{w}) = \sum_{i=1}^{N} \left(f_{\mathbf{w}}(\mathbf{p}(\mathbf{x}_i)) - y_i \right)^2 + \lambda \sum_i {w_i}^2$$

L2 regularization:
$\sum_i {w_i}^2 = \|\mathbf{w}\|_2^2$,
squared L2 norm of $\mathbf{w}$

$$= (\mathbf{P}\mathbf{w} - \mathbf{y})^T (\mathbf{P}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w}$$

人为增加

$\lambda \uparrow$ $w \downarrow$

(Hyperparameter)

$\lambda > 0$ is a **complexity/regularization parameter** that controls the amount of shrinkage: the larger the value of λ, the greater the amount of shrinkage.

Ridge Regression

➢ Learning

$$\underset{\mathbf{w}}{\operatorname{argmin}}(\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y})+\lambda\mathbf{w}^T\mathbf{w}$$

Solution: $\dfrac{dJ(\mathbf{w})}{d\mathbf{w}}=\mathbf{0}$

$$\frac{d\left((\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y})+\lambda\mathbf{w}^T\mathbf{w}\right)}{d\mathbf{w}}=\mathbf{0}$$

$$2\mathbf{P}^T\mathbf{Pw}-2\mathbf{P}^T\mathbf{y}+2\lambda\mathbf{w}=\mathbf{0}$$

$$\mathbf{P}^T\mathbf{Pw}+\lambda\mathbf{w}=\mathbf{P}^T\mathbf{y}$$

$$(\mathbf{P}^T\mathbf{P}+\lambda\mathbf{I})\mathbf{w}=\mathbf{P}^T\mathbf{y}$$

Since $\mathbf{P}^T\mathbf{P}+\lambda\mathbf{I}$ is always invertible, $\mathbf{w}^*=(\mathbf{P}^T\mathbf{P}+\lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{y}$

Ridge Regression

Proof: $\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I}$ is always invertible when $\lambda>0$

Suppose $\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I}$ is not invertible.

Columns/rows in $\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I}$ are linearly dependent

There exits a vector $\mathbf{a} \neq \mathbf{0}$, s.t., $(\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})\mathbf{a}=\mathbf{0}$

$\mathbf{a}^T(\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})\mathbf{a}=0$

矛盾

$\mathbf{a}^T\mathbf{P}^T\mathbf{P}\mathbf{a} + \lambda\mathbf{a}^T\mathbf{a} = 0$

$||\mathbf{P}\mathbf{a}||^2 + \lambda||\mathbf{a}||^2 = 0$

However, $||\mathbf{P}\mathbf{a}||^2 + \lambda||\mathbf{a}||^2 = 0$ only holds if $||\mathbf{P}\mathbf{a}||^2 = ||\mathbf{a}||^2 = 0$

$\mathbf{a} = \mathbf{0}$

Columns/rows in $\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I}$ are linearly independent

Ridge Regression

➢ Learning

Constriant → Unconstriant

Geometric Interpretation of Ridge Regression Solution

$$\underset{\mathbf{w}}{\operatorname{argmin}}(\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y}) + \lambda\mathbf{w}^T\mathbf{w}$$

Dual form

$$\underset{\boldsymbol{x}}{\operatorname{argmin}}\, f(\boldsymbol{x}) + \lambda g(\boldsymbol{x})$$

转化

Lagrangian Duality

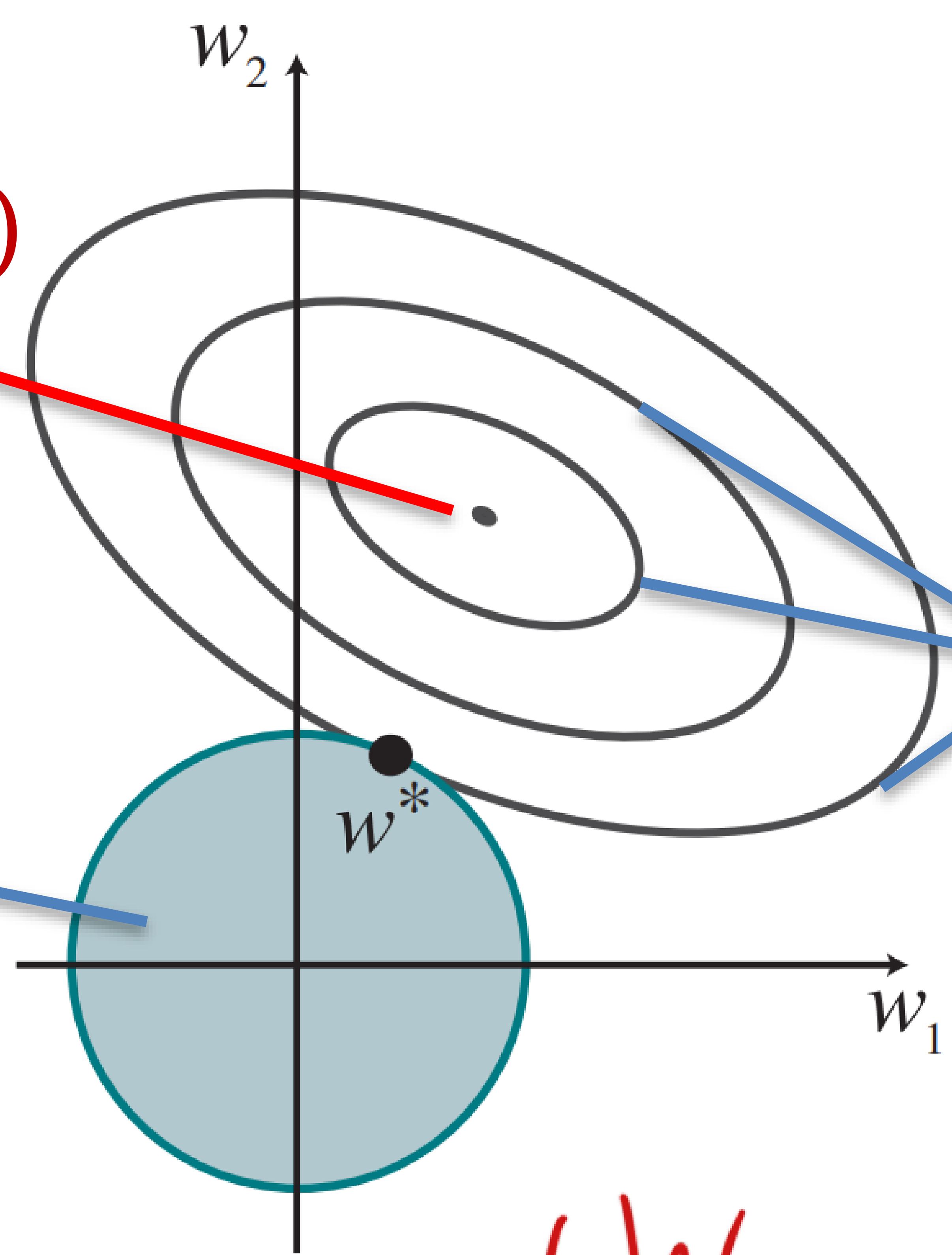
$$\underset{\mathbf{w}}{\operatorname{argmin}}(\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y}), \text{ subject to } \mathbf{w}^T\mathbf{w} \le c$$

Primal form

$$\underset{\boldsymbol{x}}{\operatorname{argmin}}\, f(\boldsymbol{x}), s.t.\, g(\boldsymbol{x}) \le 0$$

$\mathbf{w}^T\mathbf{w} - C$

$$\underset{\mathbf{w}}{\operatorname{argmin}}(\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y})$$



Set of points **w** in 2D weight space that have $\mathbf{w}^T\mathbf{w} \le c$

Contours of $(\mathbf{Pw}-\mathbf{y})^T(\mathbf{Pw}-\mathbf{y})$

C is omitted

(We take derivative to get the

Ridge Regression

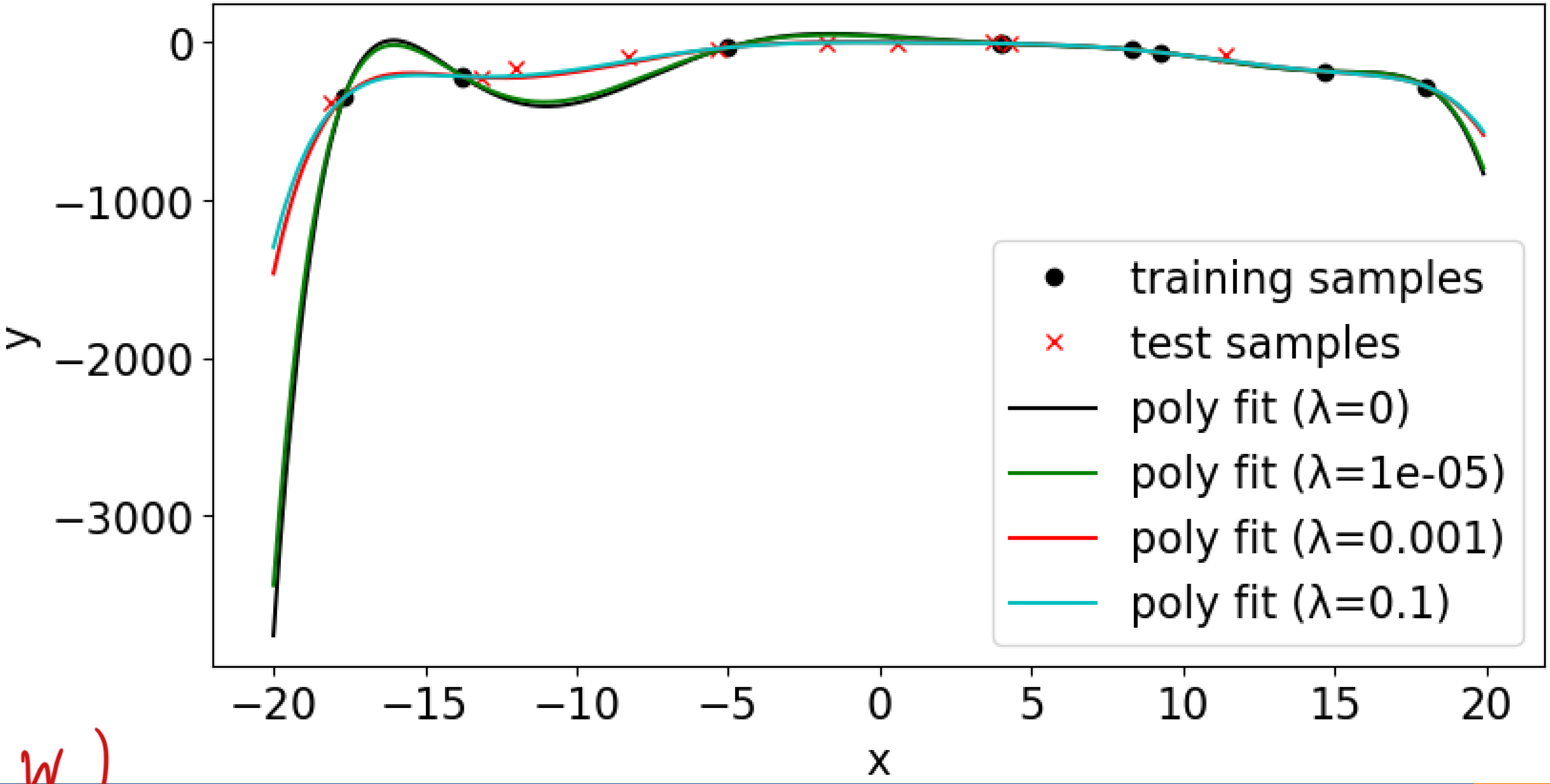
λ ↗ Flatten ↗

➢ Prediction

$$f_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{w}^*$$

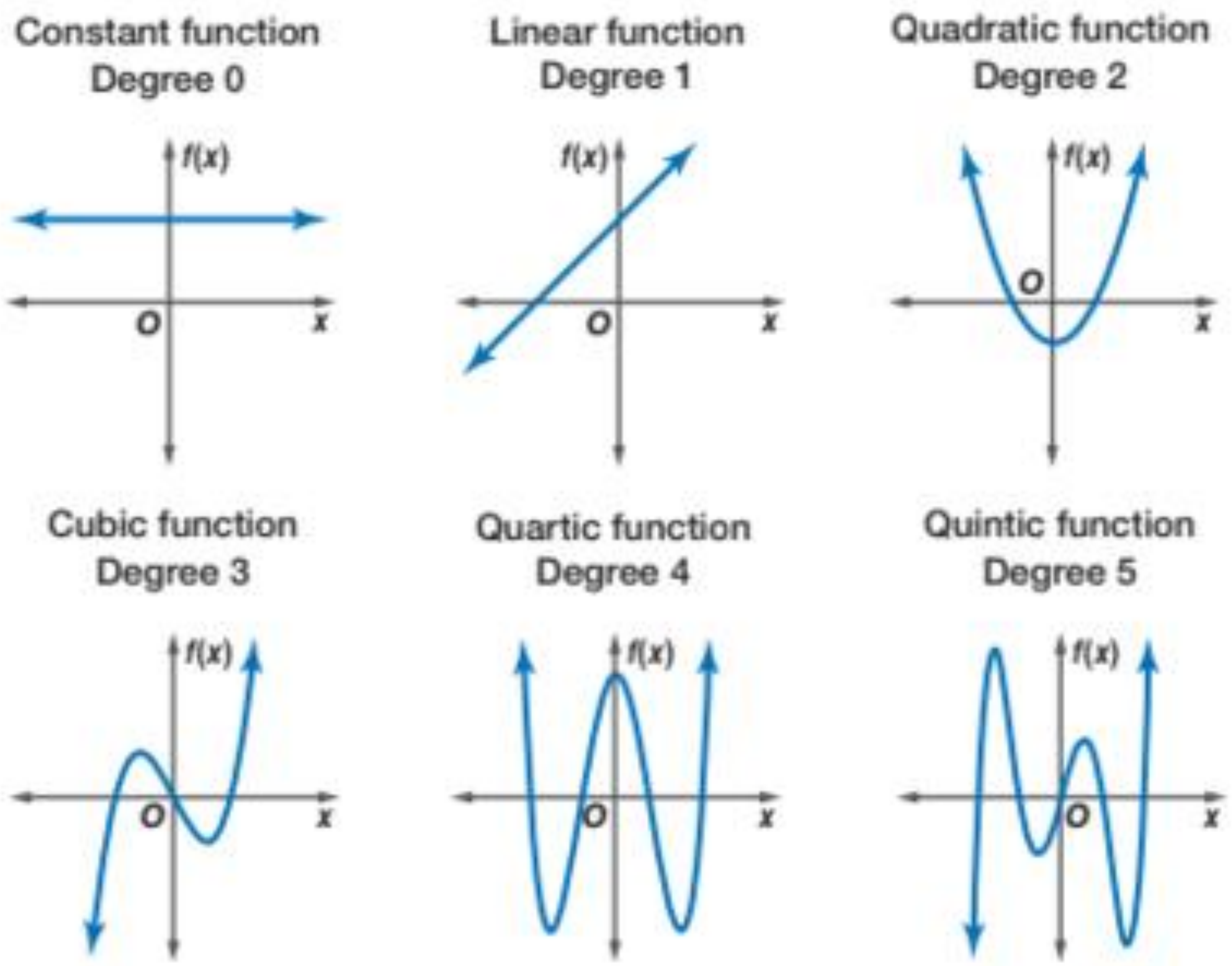
How is weight shrinkage connected with $\mathbf{P}^T\mathbf{P}$ being not invertible?

Example: Order = 8, no. of input variable =1



optimal w

Polynomial model with different degrees



Flattening
(Fewer ups and downs)
≈
Reducing degree
(Reduce no. polynomial features)

When is $\mathbf{P}^T\mathbf{P}$ not invertible? When $\boxed{\text{rank}(\mathbf{P})}$ < no. polynomial features

Rank($\mathbf{P}$)=Rank($\mathbf{P}^T\mathbf{P}$)

Can be proved by **rank-nullity theorem**

(Rank(A)+nullity(A)= No. cols)

Intrinsic dimension of the feature space

No. of independent columns.

Prove Rank(P)=Rank($\mathbf{P}^T\mathbf{P}$) by rank-nullity theorem

$\Updownarrow$

Prove Nullity($\mathbf{P}$)=Nullity($\mathbf{P}^T\mathbf{P}$)

$\Updownarrow$

Prove

- if $\mathbf{x}$ satisfies $\mathbf{Px} = \mathbf{0}$, $\mathbf{P}^T\mathbf{Px} = \mathbf{0}$

  and

- if $\mathbf{x}$ satisfies $\boxed{\mathbf{P}^T\mathbf{Px} = \mathbf{0}}$, $\mathbf{Px} = \mathbf{0}$

$$\mathbf{x}^T\mathbf{P}^T\mathbf{Px} = 0$$

$$\Downarrow$$

$$||\mathbf{Px}||^2 = 0$$

$$\Downarrow$$

$$\mathbf{Px} = \mathbf{0}$$

Ridge Regression

➢ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \ldots, (\mathbf{x}_N, \mathbf{y}_N)$,

where $\mathbf{x}$ is a d-dimensional feature vector

$\mathbf{y}$ is a h-dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f_W}(\mathbf{x}) = \mathbf{f_W}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T\mathbf{W}$,

-Learning:

Find the optimal values for $\mathbf{W}^*$ which minimizes:

$$J(\mathbf{W}) = \sum_{k=1}^{h} \left(\sum_{i=1}^{N} \left(f_{\mathbf{w}_k}(\mathbf{p}(\mathbf{x}_i)) - y_{i,k} \right)^2 + \lambda \sum_{i} w_{i,k}{}^2 \right)$$

$$= \sum_{k=1}^{h} \left((\mathbf{P}\mathbf{w}_k - \mathbf{y}_k)^T (\mathbf{P}\mathbf{w}_k - \mathbf{y}_k) + \lambda \mathbf{w}_k{}^T \mathbf{w}_k \right)$$

$$\mathbf{Y} = \begin{bmatrix} y_{1,1} & \cdots & y_{1,h} \\ \vdots & \ddots & \vdots \\ y_{N,1} & \cdots & y_{N,h} \end{bmatrix}$$

Solution in Compact Form: $\mathbf{W}^* = (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{Y}$

Ridge Regression

➢ Extend to Multiple Outputs

Input: $(\mathbf{x}_1, \mathbf{y}_1), (\mathbf{x}_2, \mathbf{y}_2), \dots, (\mathbf{x}_N, \mathbf{y}_N)$,

where $\mathbf{x}$ is a d-dimensional feature vector

$\mathbf{y}$ is a h-dimensional target vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}, \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_h \end{bmatrix}$$

Output: $\mathbf{f}_\mathbf{W}(\mathbf{x}) = \mathbf{f}_\mathbf{W}(\mathbf{p}(\mathbf{x})) = \mathbf{p}^T\mathbf{W}$,

-Learning: $\mathbf{W}^* = (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{W}^*$

Evaluation Metrics for Regression

Evaluation Metrics for Regression

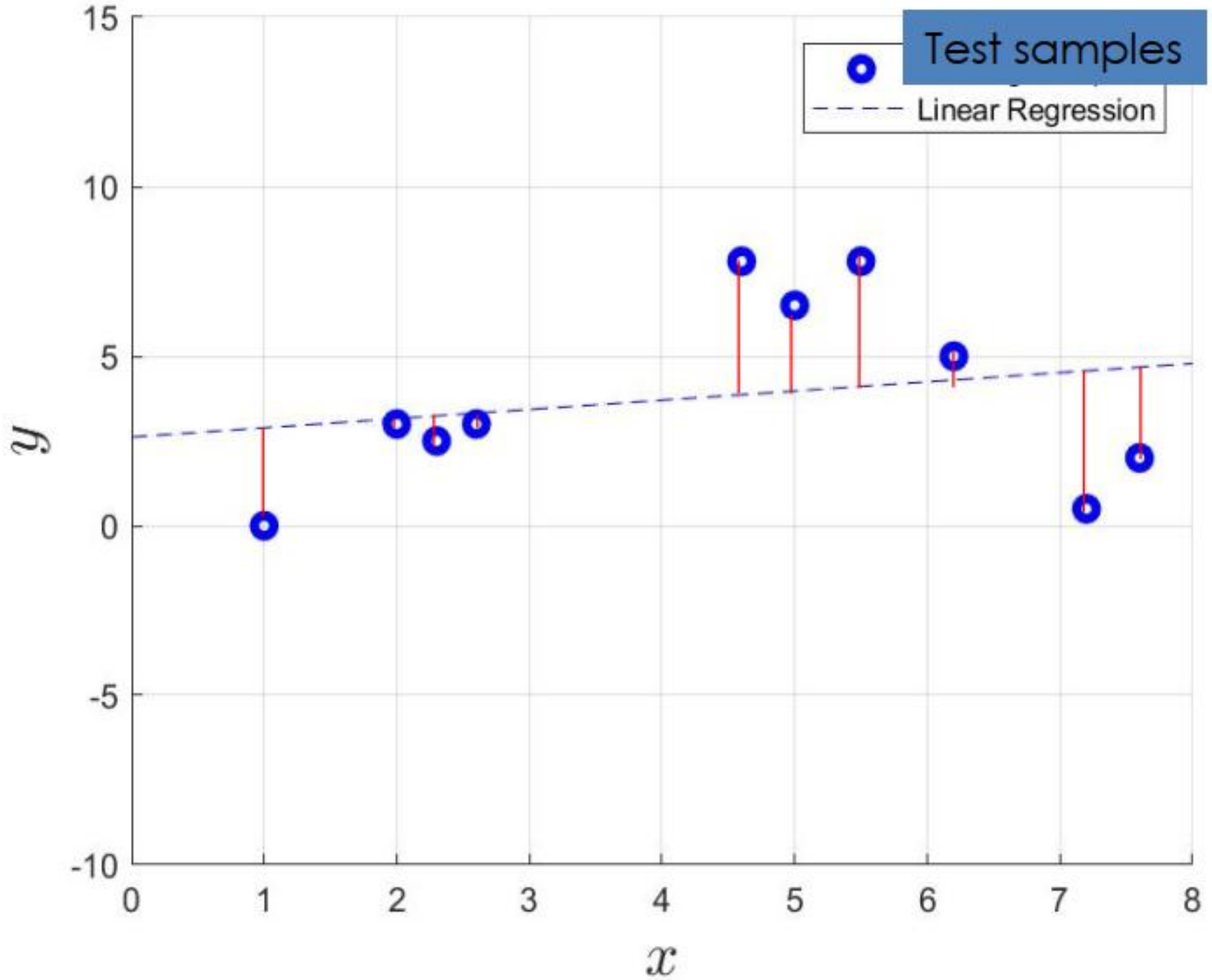
Mean Square Error

$$(\text{MSE} = \frac{\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2}{n})$$

Mean Absolute Error

$$(\text{MAE} = \frac{\Sigma_{i=1}^{n}|y_i - \hat{y}_i|}{n})$$

where y_i denotes the target output and $\hat{y}_i$ denotes the predicted output for sample i.




```
from sklearn.metrics import mean_absolute_error, mean_squared_error
```

```
mae = mean_absolute_error(y_true, y_pred)
mse = mean_squared_error(y_true, y_pred)
```

Summary

✓ **Linear Regression**

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{X}_t) = \mathbf{X}_t\mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{X}_t) = \mathbf{X}_t\mathbf{W}^*$

✓ **Polynomial Regression**

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{P}^T\mathbf{P})^{-1}\mathbf{P}^T\mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{W}^*$

✓ **Ridge Regression**

Scalar output for each sample

-Learning: $\mathbf{w}^* = (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{y}$

-Prediction: $\mathbf{f}_{\mathbf{w}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{w}^*$

Vector output for each sample

-Learning: $\mathbf{W}^* = (\mathbf{P}^T\mathbf{P} + \lambda\mathbf{I})^{-1}\mathbf{P}^T\mathbf{Y}$

-Prediction: $\mathbf{F}_{\mathbf{W}^*}(\mathbf{P}(\mathbf{X}_t)) = \mathbf{P}_t\mathbf{W}^*$

✓ **2 Evaluation Metrics for Regression Task**